

Lab 3: Automatic Speech Recognition and Translation Report

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1 Toronto ASR — Whisper Fine-tuning

1.1 Data overview

The Toronto corpus is a small Ukrainian read-speech dataset of ~ 170 utterances paired with transcripts in `labels.jsonl`. Audio is single-channel and was resampled to 16 kHz for Whisper. Utterance length ranges from a few seconds to ~ 30 s, comfortably inside Whisper’s 30-second receptive field.

Main features and problems.

- **Size.** With ~ 170 clips and 21 held out for testing, the train+val pool is roughly 150 clips — borderline for adaptation, and risky for overfitting and vocabulary undercoverage.
- **Speaker / domain narrowness.** A handful of speakers and one recording style; the model can memorize prosody rather than learn Ukrainian phonotactics.
- **Orthographic edge cases.** Apostrophes, hyphenation and punctuation are inconsistent across labels; the Whisper Ukrainian tokenizer splits these into many sub-tokens, inflating WER even when the phonetic output is correct.
- **Test split.** The 21 `test_lines` listed in the brief are used only for final scoring — never for training or validation — to avoid leakage.

1.2 Approach

Backbone. `openai/whisper-base` was chosen as the pretrain. It already has a Ukrainian tokenizer and produces sensible (if noisy) Ukrainian out of the box; larger Whisper variants were rejected to avoid rapid overfitting on this dataset.

Training stack. PyTorch Lightning with `WhisperForConditionalGeneration`, `WhisperProcessor`, and a custom `TorontoWhisperDataModule` that handles resampling, log-Mel feature extraction, label tokenization, and -100 masking on padding. Mixed precision, gradient clipping, and `ModelCheckpoint` on `val_wer` were used.

Experiments. Three configurations were compared: a zero-shot baseline of `whisper-base`, a fine-tune with the encoder frozen (decoder-only adaptation, $\text{lr } 1 \times 10^{-5}$), and a fully-trainable fine-tune with the encoder unfrozen ($\text{lr } 5 \times 10^{-6}$).

1.3 Metrics

Numbers below are the end-of-training values from the Lightning CSV logs (`csv_logs/whisper-base-uk/`). The zero-shot baseline was scored on a stratified sample of 210 clips (10 per held-out speaker).

1.4 Hypotheses and discussion

- **Frozen-encoder fine-tuning barely beats zero-shot.** The CER is essentially unchanged ($28.79 \rightarrow 28.56$); freezing the encoder lets the decoder memorize transcript patterns but does not teach the model the acoustics of Toronto speakers.

Table 1: ASR performance on the Toronto held-out test set ($\dagger$ 210-clip stratified sample for the baseline).

Configuration	WER $\downarrow$	CER $\downarrow$
Whisper-base, zero-shot (no fine-tune) †	62.00	28.79
Fine-tune, encoder frozen, lr 1×10^{-5}	59.02	28.56
Fine-tune, encoder unfrozen, lr 5×10^{-6}	53.83	25.19

- **Unfreezing the encoder moves the needle.** Going fully-trainable cuts WER by another ~ 5 absolute points ($59.02 \rightarrow 53.83$) and CER by ~ 3 ($28.56 \rightarrow 25.19$). A low LR (5×10^{-6}) keeps the pretrained features from collapsing.
- **WER overstates the real error.** CER is roughly half of WER — a single wrong morpheme or missed apostrophe flips an entire token, but the surrounding characters are often correct.
- **Dataset size is the binding constraint.** Both runs plateau quickly and train loss falls much faster than val loss — the classic small-data overfit signature.

2 AST Level 1 — Basic Fine-Tuning on FLEURS

This task demonstrates an end-to-end EN-audio $\rightarrow$ UK-text training pipeline on a small existing parallel dataset. Implementation: `lab3_additional.ipynb`.

2.1 Data overview

Alignment was performed using the `google/fleurs` corpus by inner-joining the `en_us` audio side with the `uk_ua` text side by `id`.

Table 2: FLEURS EN $\rightarrow$ UK aligned pair counts.

Split	Aligned pairs
train	1,425
validation	138
test	329

2.2 Approach

Backbone. `openai/whisper-small` (241 M parameters) in mixed precision.

Training stack. PyTorch Lightning with `FleursASTDataset`, `WhisperDataCollator` (masking pad tokens with -100), and `WhisperASTLightningModule`. Hyper-parameters: lr 1×10^{-5} , weight-decay 1×10^{-2} , batch size 2.

Two task framings. Run A uses Whisper’s `task="transcribe"` with Ukrainian output; Run B uses `task="translate"` with Ukrainian as the target.

2.3 Metrics

Qualitative sample (Run A, same clip in Run B).

Table 3: Level 1 AST results on the FLEURS test set (329 clips).

Configuration	Val loss ↓	COMET ↑
Run A — <code>task=transcribe</code> , 5 epochs	2.4099	0.3594
Run B — <code>task=translate</code> , 8 epochs	2.3810	0.3581

SRC: *he did not set a figure for the cuts saying they will be made based on china's economic output*

REF: *він не встановив показник скорочень сказавши що це буде залежати від економічного обсягу виробництва китаю*

HYP (A): *він не розказувався зі створенням що вони будуть створені . . . інформаційний інформаційний інформаційний. . .*

HYP (B): *він не встановив зручну фіктуру на кордоні стверджувача. . . економічної випадки економічної. . .*

3 AST Level 2 — LibriSpeech-UA Dataset and Fine-tune

This task constructs a *new* EN-audio → UK-text parallel dataset and demonstrates objective improvement over the baseline.

3.1 Data overview

Source corpus. LibriSpeech ASR (`train.clean.100`). Ukrainian targets were produced using `nllb-200-distilled-600M`. A novelty check confirmed no public Ukrainian translation of LibriSpeech transcripts exists.

Table 4: LibriSpeech-UA split sizes.

Split	Records
train	28,539
validation.clean	2,703
test.clean	2,620
test.other	2,939

3.2 Approach

We fine-tuned `whisper-base` using `Seq2SeqTrainer`. Decoder conditioning forces `<|en|><|transcribe|>` while labels are tokenized in Ukrainian. This repurposes the English acoustic encoder for Ukrainian output.

3.3 Metrics

Table 5: COMET (Unbabel/wmt22-comet-da) on LibriSpeech-UA test splits.

Split	Records	Baseline	Fine-tuned	Δ
test.clean	2,620	0.3209	0.4174	+0.0965
test.other	2,939	0.3266	0.4139	+0.0873

3.4 Conclusion

Fine-tuning on the new LibriSpeech-UA dataset improves COMET by **+0.087–0.097** absolute. The dataset and artifacts serve as a contribution to open Ukrainian Audio ML.